Review notes · for you

Predict with data: unit check

Data science · Level 3 · Unit check

Time	Materials
25 min	Paper and a pen. No device needed.

Learning goals

- Lines that predict
- Training and testing
- Nearest neighbours
- Judging a model

Activity: Revise in pairs

1. Pairs (2 min): Work with a partner and share one sheet for your answers.
2. Quiz (15 min): Read each question, agree on an answer and say why.
3. Check (8 min): Compare with the answers and redo the ones you got wrong.

Discussion

- Which idea will you use first?
- Which school model would you question?
- Which mistake costs most at your school?

Check yourself

- One prediction my class could test fairly this week.

Try it this week

Make a prediction about something in your week, test it fairly, and judge it by its mistakes.

Quiz sheet

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Name _____ Class _____ Date _____

1 **Use y = 6 + 3x.**

At x = 2 the prediction is _____.

Word bank: 9 · 12

2 **Predicting far outside the data's range is extrapolation.**

◯ True ◯ False

3 **Which score estimates performance on new data?**

◯ The best score after many tries on the test set ◯ The training score ◯ The test score

4 **Match each term to its meaning.**

Overfitting	• • • •	Share of real positives caught
Recall	• • • •	Share of flags that were right
Precision	• • • •	Low train error, high test error

5 **k = 5 neighbours: A, B, A, A, B. Prediction?**

◯ A ◯ B ◯ A tie

6 **k-NN features should be on comparable scales.**

◯ True ◯ False

7 **TP = 4, FP = 4, FN = 1.**

Precision is _____ and recall is _____.

Word bank: 1 · 0.5 · 0.8 · 0.4

8 **Lowering the threshold usually… Choose all that apply.**

☐ raises false positives ☐ raises precision ☐ raises recall

9 **Order the workflow.**

☐ Test once

☐ Split

☐ Train

10 **A regression line shows cause and effect.**

◯ True ◯ False

Check your answers

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1. **At x = 2 the prediction is 12.**
   6 + 3 × 2 = 12.

2. **True**
   And it has no evidence behind it.

3. **The test score**
   Only an untouched test set is honest.

4. **Overfitting — Low train error, high test error**
   Recall — Share of real positives caught
   Precision — Share of flags that were right
   Key evaluation ideas.

5. **A**
   A wins three votes to two.

6. **True**
   Otherwise the feature with the widest range, like weight in grams, dominates distances.

7. **Precision is 0.5 and recall is 0.8.**
   4 ÷ 8 = 0.5 and 4 ÷ 5 = 0.8.

8. **raises recall; raises false positives**
   More flags: more catches and more false alarms.

9. **Top to bottom: 3, 1, 2**
   Split before you train.

10. **False**
    It shows association only.